



AMERICAN SCHOOL OF DUBAI

AP Calculus BC SUMMER Assignment

NAME: _____

PRECALCULUS REVIEW

Solve the following inequalities. Answers should be in interval notation form.

1.) $-x(x-2)^2(x+3) > 0$

Solve for x.

3.) $2^{x-1} - 32 = 0$

4.) $e^{2x} - 2e^x = 3$

5.) Find the equations of the lines passing through $(-2,5)$ and having the following characteristics.

(a) slope of $\frac{-2}{3}$

(c) Passing through the point $\left(\frac{1}{3}, -1\right)$

(b) Perpendicular to the line $-2x + y = 1$

(d) Parallel to the x-axis

6.) Given $f(x) = \frac{1 - 3x + 2x^2}{4 - 9x^2}$. Find the following:

a.) x-intercept(s)

c.) vertical asymptote(s)

b.) y-intercept(s)

d.) horizontal asymptote(s)

7.) Evaluate (if possible) the function at each value of the independent variable.

$$f(x) = \begin{cases} 1 - 2x^2, & x < 0 \\ |x - 2|, & x \geq 0 \end{cases}$$

(a) $f\left(\frac{1}{2}\right)$

(b) $f(0)$

(c) $\frac{f(3) - f(-3)}{3 - (-3)}$

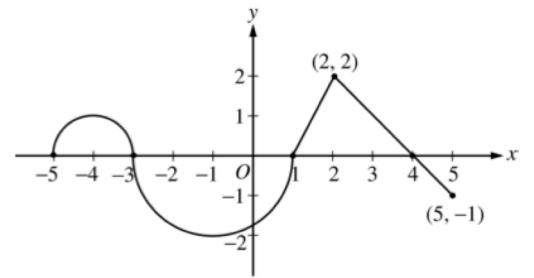
8.) Write an algebraic expression that is equivalent to the expression $\cos\left(\arcsin\left(\frac{x}{2}\right)\right)$.

9.) Find the inverse of the following functions. Then determine the domain and range of the given functions and their inverse.

a) $g(x) = \frac{1}{3}e^{x-2}$

b.) $h(x) = \frac{x-2}{x+1}$

10.) Find the area of the two semicircles (from -5 to 1) and two triangles (from 1 to 5).



11.) If $y = \arcsin x$, then $\cos y =$ _____.

- (A) $\sqrt{1-x^2}$ (B) $\sqrt{x^2-1}$ (C) $\frac{1}{\sqrt{x^2-1}}$ (D) $\frac{1}{\sqrt{1-x^2}}$ (E) $\frac{1}{x}$

12.) Find the sum of the given series: $6 + 3 + \frac{3}{2} + \frac{3}{4} + \dots$

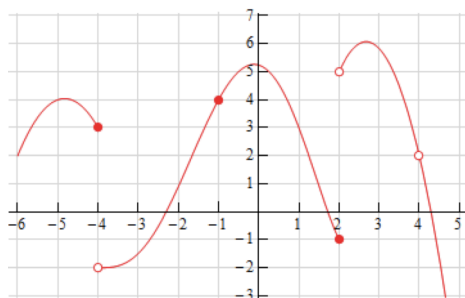
13.) Find the value(s) of θ for which the given polar graphs intersect.

$$r = 3$$

$$r = 6\sin(2\theta)$$

14.) Write the partial fraction decomposition of $R(x) = \frac{7-x}{x^2-2x-3}$.

LIMITS

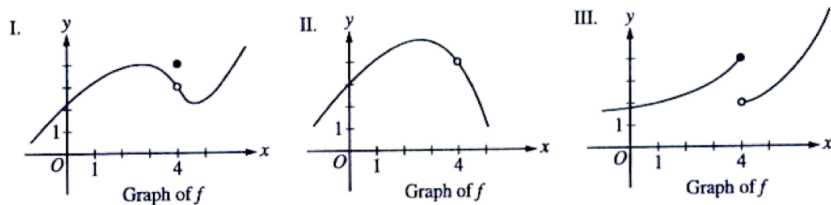


1) The graph of $f(x)$ is given above. Evaluate the following limits:

a.) $\lim_{x \rightarrow -4} f(x) =$ b.) $\lim_{x \rightarrow -1} f(x) =$ c.) $\lim_{x \rightarrow 2} f(x) =$ d.) $\lim_{x \rightarrow 4} f(x) =$

2) For which of the following does $\lim_{x \rightarrow 4} f(x)$ exist?

(A) I only (B) II only (C) III only (D) I and II only (E) I and III only



3) Evaluate the following limits:

a.) $\lim_{x \rightarrow -2} 3 =$ b.) $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x - 2} =$ c.) $\lim_{x \rightarrow -1} \frac{x + 1}{x^3 + 1} =$

4) Evaluate the following limits:

a.) $\lim_{x \rightarrow 1} \frac{\sqrt{x} - 1}{x - 1} =$

b.) $\lim_{x \rightarrow 1} \frac{\frac{1}{x} - 1}{2x - 2} =$

5) Evaluate $\lim_{x \rightarrow 0} \frac{1 - \cos x - 2x}{x^2 - 2x}$

6) $\lim_{x \rightarrow 0} \frac{\sin 4x}{x} =$

7) Evaluate the following limits:

a.) $\lim_{x \rightarrow 0} \frac{1 - \cos^2(2x)}{x^2} =$

b.) $\lim_{x \rightarrow 0} \frac{x^2 - 3x}{\tan x} =$

8) Evaluate the following limits:

a.) $\lim_{x \rightarrow 1^-} \frac{2x+1}{3x-7} =$

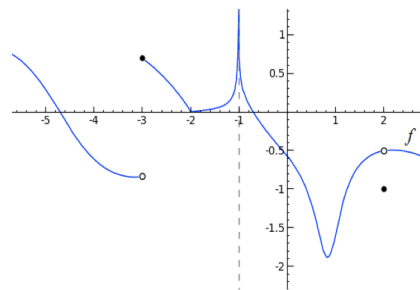
b.) $\lim_{x \rightarrow -3^+} \sqrt{1+2x} =$

c.) $\lim_{x \rightarrow 2^-} \frac{4-x^2}{8-x^3} =$

9) Given $f(x) = \begin{cases} 2x+4, & x \leq 2 \\ 1-x^2, & x > 2 \end{cases}$, then $\lim_{x \rightarrow 2^+} f(x) =$.

10) The graph of $f(x)$ is given to the right.

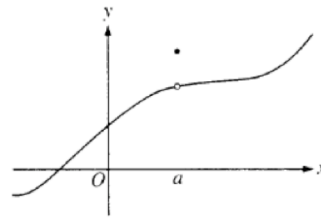
Evaluate $\lim_{x \rightarrow -3^-} f(x) =$



CONTINUITY

1) The graph of the function f is shown in the figure below. Which of the following statements about f is false?

- (A) $f(a)$ is defined
- (B) $\lim_{x \rightarrow a} f(x)$ exists
- (C) $\lim_{x \rightarrow a^-} f(x)$ is equal to $\lim_{x \rightarrow a^+} f(x)$
- (D) $x = a$ is in the domain of f
- (E) f is continuous at $x = a$



2) Find the constants a and b such that the function is continuous on the entire real line.

$$f(x) = \begin{cases} 2, & x \leq -1 \\ ax + b, & -1 < x < 3 \\ -2, & x \geq 3 \end{cases}$$

3) Find the value of k so that the function $g(x)$ is continuous everywhere.

$$g(x) = \begin{cases} \frac{x-3}{x^2-9}, & x \neq 3 \\ k, & x = 3 \end{cases}$$

4) Hot water is dripping through a coffeemaker, filling a large cup with coffee. The amount of coffee in the cup at time t , $0 \leq t \leq 6$, is given by a continuous function C , where t is measured in minutes. Selected values of $C(t)$, measured in ounces, are given in the table below.

t (minutes)	0	1	2	3	4	5	6
$C(t)$ (ounces)	0	5.3	8.8	11.2	12.8	13.8	14.5

Is there a time t , $2 \leq t \leq 4$, at which $C(t) = 10$? Justify your answer.

5) Evaluate the following limits:

a.) $\lim_{x \rightarrow 1^-} \frac{x-2}{x-1}$

b.) $\lim_{x \rightarrow 1^+} \frac{x+2}{1-x}$

c.) $\lim_{x \rightarrow \left(\frac{\pi}{2}\right)^-} \tan x$

6) Evaluate the following limits:

a.) $\lim_{x \rightarrow +\infty} \frac{2x-3}{3x+1}$

b.) $\lim_{x \rightarrow -\infty} \frac{2x-3}{3x+1}$

c.) $\lim_{x \rightarrow -\infty} \frac{4x+3x^2}{5x-6x^2}$

7) $\lim_{x \rightarrow \infty} \frac{(2x-1)(3-x)}{(x-1)(x+3)} =$

(A) -3

(B) -2

(C) 2

(D) 3

(E) Does Not Exist

8) Given $f(x) = \frac{4x+3}{\sqrt{9x^2+1}}$. Is $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow -\infty} f(x)$? Explain.

9) $\lim_{n \rightarrow \infty} \frac{3n^3 - 5n}{n^3 - 2n^2 + 1} =$

(A) -5

(B) -2

(C) 1

(D) 3

(E) Does Not Exist

10) If $a \neq 0$, then $\lim_{x \rightarrow a} \frac{x^2 - a^2}{x^4 - a^4}$ is _____.

(A) $\frac{1}{a^2}$

(B) $\frac{1}{2a^2}$

(C) $\frac{1}{6a^2}$

(D) 0

(E) Does Not Exist

11.) For what value of k is the function $f(x) = \begin{cases} 2x - kx^2, & x \leq 3 \\ k + 3x, & x > 3 \end{cases}$ continuous at $x = 3$? Show the work that leads to your answer.

12.) A function f is continuous on $[0,5]$ and some of the values of f are shown below. If $f(x) = -2$ has no solution on $[0,5]$, then b could be _____.

x	0	3	5
f	- 4	b	- 4

(A) 3

(B) 1

(C) 0

(D) - 2

(E) - 5

Differentiation

1) $\lim_{\Delta x \rightarrow 0} \frac{\cos^2(x + \Delta x) - \cos^2 x}{\Delta x} = \underline{\hspace{2cm}}$

2) $\lim_{h \rightarrow 0} \frac{\sqrt{4+h} - \sqrt{4}}{h} = \underline{\hspace{2cm}}$

3) Given $\lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = 2$. Which of the following functions cannot be $f(x)$?

(A) $y = 2x + 1$ (B) $y = x^2 + 3$ (C) $y = x^3 - x^2$ (D) $y = 4\sqrt{x}$ (E) $y = \frac{1}{\pi} \sin(\pi x) + 3x$

4) Is the function $f(x) = \begin{cases} x^2, & x < 1 \\ 2x, & x \geq 1 \end{cases}$ differentiable at $x = 1$? Explain your reasoning.

5) Which of the following functions is differentiable at $x = 1$?

- (A) $f(x) = |x - 1|$ (B) $g(x) = \frac{1}{x - 1}$ (C) $h(x) = \sqrt{x - 1}$ (D) $j(x) = (x - 1)^2$ (E) $k(x) = \frac{|x - 1|}{x - 1}$

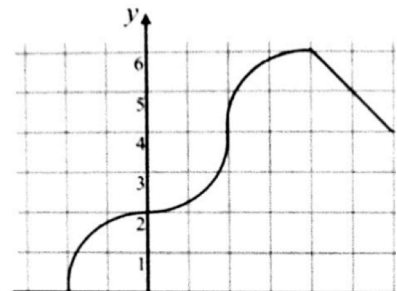
6) If $\lim_{x \rightarrow 5} \frac{f(x) - f(5)}{x - 5} = 3$, then which of the following must be true?

- I. $\lim_{x \rightarrow 5} f(x) = 3$
- II. $f(x)$ is continuous at $x = 5$.
- III. $f(x)$ is differentiable at $x = 5$.

- (A) I only (B) II only (C) III only (D) II and III only (E) I, II, and III

7) The graph of $y = f(x)$ below consists of three quarter circles and a line segment. For which values of x in the open interval $(-2, 6)$ is f not differentiable?

- (A) 0 only (C) 0 and 2 only (E) 0, 2, and 4
(B) 2 only (D) 2 and 4 only



8) Find a and b such that $f(x)$ is differentiable everywhere.

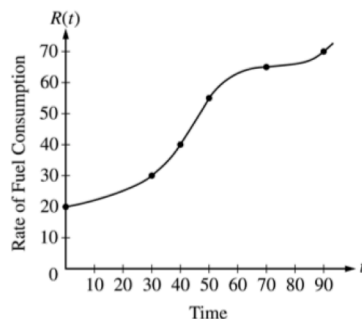
$$f(x) = \begin{cases} ax^2 + x, & x \leq 1 \\ 2x^3 + bx, & x > 1 \end{cases}$$

9) The average rate of change of $f(x) = 4x - x^2$ on the interval $[1, 3]$ is _____.

- (A) 2 (B) 1 (C) 0 (D) - 1 (E) 2

10) The rate of fuel consumption, in gallons per minute, recorded during an airplane flight is given by a twice-differentiable and strictly increasing function R of time t . The graph of R and a table of selected values of $R(t)$, for the time interval $0 \leq t \leq 90$ minutes, are shown below.

(a) Use the data from the table to find an approximation for $R'(80)$. Show the computations that lead to your answer. Indicate units of measure.



t (minutes)	$R(t)$ (gallons per minute)
0	20
30	30
40	40
50	55
70	65
90	70

11) Determine the instantaneous rate of change of the function $f(x) = 5e^{\frac{1}{2}x^2}$ at $x = 2$. (CALCULATOR)

12) If $f(x) = x^{\frac{3}{2}}$, then $f'(4) = \underline{\hspace{2cm}}$.

(A) - 6

(B) - 3

(C) 3

(D) 6

(E) 8

13) If $f(x) = (x-1)^2 \sin x$, then $f'(0) = \underline{\hspace{2cm}}$.

(A) - 2

(B) - 1

(C) 0

(D) 1

(E) 2

14) Find $\frac{d^2y}{dx^2}$ if $y = x \cos x$.

15) If $f(x) = x^3 - x^2 + \frac{1}{x}$, which of the following is $f''(2)$?

(A) $\frac{31}{4}$

(B) $\frac{39}{4}$

(C) $\frac{41}{4}$

(D) $\frac{79}{8}$

(E) $\frac{81}{8}$

16) $\frac{d}{dx}[\cos^2(x^3)] = \underline{\hspace{2cm}}$.

(A) $\sin^2(x^3)$

(C) $-2\sin(x^3)\cos(x^3)$

(E) $6x^2\sin(x^3)\cos(x^3)$

(B) $6x^2\cos(x^3)$

(D) $-6x^2\sin(x^3)\cos(x^3)$

17) If $f(x) = \cot^2(2x)$, then $f'\left(\frac{\pi}{6}\right) = \underline{\hspace{2cm}}$.

18) Find $\frac{d^2y}{dx^2}$. $x^2 - y^3 = 2y$

19) Find $\frac{dy}{dx}$ and evaluate at the given point. $(x+y)^3 = x^3 + y^3, \quad (-1,1)$

21) A particle moves along the x-axis with velocity at time $t \geq 0$ given by $v(t) = -1 + e^{1-t}$.

a) Find the acceleration of the particle at time $t = 3$.

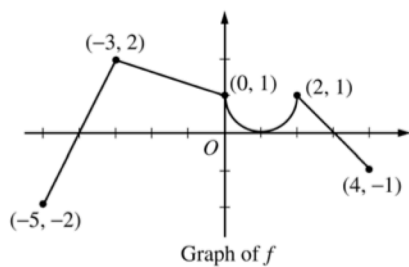
b) Is the speed of the particle increasing at time $t = 3$? Give a reason for your answer.

c) Find all values of t at which the particle changes direction. Give a reason for your answer.

d) Find the total distance traveled by the particle over the time interval $0 \leq t \leq 3$. *

(CALCULATOR)

- Is the level of heating oil in the tank rising or falling at time $t = 6$ hours? Give a reason for your answer.
- How many gallons of heating oil are pumped into the tank during the time interval $0 \leq t \leq 12$ hours?
- How many gallons of heating oil are in the tank at time $t = 12$ hours?
- At what time, t , for $0 \leq t \leq 12$, is the volume of heating oil in the tank the least? Show the analysis that leads to your answer. *



23) The graph of the function f shown above consists of a semicircle and three line segments. Let $g'(x) = f(x)$.

- a) Find $g'(0)$ and $g''(0)$.

- b) Find all values of x in the open interval $(-5, 4)$ at which g attains a relative maximum. Explain.

- c) Find the absolute minimum value of g on the closed interval $[-5, 4]$. Justify your answer.

- d) Find all values of x in the open interval $(-5, 4)$ at which the graph of g has a point of inflection.

II. Circle the letter of the correct answer. Explain your reasoning.

24.) Let f be a function on the closed interval $[-2,4]$. If $f(-2)=f(4)$, then there is at least one number c in $(-2,4)$ such that $f'(c) = 0$.

A. TRUE

B. FALSE

25.) Rolle's Theorem applies to the function $f(x) = x^2 + 4x - 5$ on the interval $[-4,0]$.

A. TRUE

B. FALSE

26.) On the closed interval $[0,3]$, which of the following functions can the Mean Value Theorem be applied to?

I. $f(x) = \left(x - \frac{3}{2}\right)^2$

II. $g(x) = \sqrt{x-2}$

III. $h(x) = |x-1|$

A. I only

B. II only

C. III only

D. I and II only

E. II and III only

27.) A function $f(x)$ exists such that $f'(x) = x^2 - 4$. Find all value(s) of x for which f is increasing.

(A) $(-\infty, -2)$

(B) $(2, \infty)$

(C) $(-\infty, -4)$

(D) $(-2, 2)$

(E) $(-\infty, -2) \cup (2, \infty)$

28.) The derivative of $g(x)$ exists and is equal to $g'(x) = (x-2)(x+1)(x-3)^2$. Find all value(s) of x such that g has a relative extrema?

(A) $x = -1$ only

(B) $x = 2$

only

(C) $x = 3$ only

(D) $x = -1, 2$ only

(E) $x = -1, 2, 3$ only